Targeted Dephasing as a Causal Handle on Local Occupation Loss in an Open Bose–Hubbard Chain

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We identify a structured, three-regime causal-handle landscape in a one-dimensional open Bose–Hubbard chain under Lindblad dephasing. Sites of high local occupation variance $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$ — which, by the reflection symmetry of the model, coincide with the geometrically central sites — are tested as causal handles: additional dephasing at the top-k highest-F_i sites is compared against matched-budget random targeting across 100 independent trials per condition. In the *negative regime* ($J/U = 0.12$), targeted intervention produces less local occupation loss than random targeting at all tested sizes and horizons. In the *positive pocket* ($J/U \gtrsim 0.24$), targeted intervention produces reliably more loss, with 95% bootstrap confidence intervals excluding zero across all tested horizons ($\tau \in \{1, 2, 3\}$) and the full tested size ladder ($L \in \{6, 7, 8, 9\}$). Between these regimes lies a *crossover band* near $J/U \approx 0.18$–0.24: the onset of positive handle effectiveness shifts upward with system size and downward with measurement horizon, both trends monotone across the ladder. The site-resolved spatial response is consistent with nonlocal redistribution of occupation change rather than local instability. All evolution is exact Lindblad time evolution in the fixed-particle-number sector; no approximations are employed. The result is strictly bounded to the tested intervention family and parameter range; no thermodynamic-limit or universal-handle claims are made.

I. INTRODUCTION

The one-dimensional Bose–Hubbard model is a standard setting for studying the interplay of quantum tunnelling and on-site interactions in interacting boson systems [1, 2]. At zero temperature the model exhibits a quantum phase transition between a superfluid phase (large J/U) and a Mott-insulating phase (small J/U), controlled by the ratio of tunnelling amplitude J to on-site interaction U [3, 4]. The clean-system phase diagram and its experimental realisation in optical lattices [5] have been extensively studied [4].

In the presence of dissipation—modelled by local dephasing in the Lindblad–Gorini–Kossakowski–Sudarshan (LGKS) framework [6–8]—the system is driven away from the ground state and develops spatially heterogeneous fluctuation structure. Open quantum systems of this type arise naturally in cold-atom experiments where coupling to photon-field fluctuations causes local dephasing of on-site number states [9, 10]. More broadly, the engineering of dissipation as a resource for state preparation and quantum control has attracted significant theoretical and experimental attention [11–14].

In classical lattice systems, local structural variables can serve as causal handles: targeted perturbations at structurally selected sites produce measurably different future outcomes from matched-budget perturbations at random sites [15]. Whether analogous causal handles exist in open quantum systems—where dynamics are governed by a Lindblad master equation and the state is a density matrix rather than a classical configuration—is an empirical question that has received little direct attention.

This paper tests one specific instance. Under uniform dephasing, sites in the Bose–Hubbard chain develop different local occupation variances $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$, reflecting the degree of local number fluctuation. We ask whether selecting the highest-F_i sites for additional dephasing produces a different pattern of future occupation loss than applying the same total dephasing budget to randomly chosen sites. The question is about the causal accessibility of future dynamical outcomes through structurally targeted interventions, not about the equilibrium phase diagram.

II. MODEL AND METHODS

A. Bose–Hubbard Hamiltonian

The Hamiltonian for a chain of L sites with open boundary conditions is

$$\hat{H} = -J \sum_{i=1}^{L-1} \left(\hat{a}_i^\dagger \hat{a}_{i+1} + \text{h.c.} \right) + \frac{U}{2} \sum_{i=1}^{L} \hat{n}_i (\hat{n}_i - 1), \quad (1)$$

where J is the tunnelling amplitude, U the on-site interaction, $\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i$, and the local Fock space is truncated at $n_{\max} = 3$ occupations per site. The system is studied at half-filling ($N = \lfloor L/2 \rfloor$ particles) in the fixed-particle-number sector. Open boundary conditions are used throughout, so that the local variance field F_i can develop nontrivial spatial structure across the chain.

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B. Hilbert space

The computation is restricted to the sector of exactly N particles distributed across L sites, each with at most $n_{\max} = 3$ occupations. The dimension of this sector is $\binom{L+N-1}{N}$ reduced by the $n_{\max}$ constraint; for $L = 6$, $N = 3$ the dimension is $D = 56$; for $L = 7$, $N = 3$ it is $D = 84$; for $L = 8$, $N = 4$ it is $D = 322$; and for $L = 9$, $N = 4$ it is $D = 486$. For the two smaller sizes ($D \leq 84$), all operators are constructed as dense matrices and time evolution is computed via the direct Padé matrix-exponential path. For the two larger sizes ($D \geq 322$), the Liouvillian superoperator (acting on the D^2-dimensional vectorised density matrix) is stored as a sparse CSR matrix; time evolution uses the Krylov-based `expm_multiply` path, and the per-trial modified Liouvillian is handled as a lightweight linear operator that avoids allocating a new sparse matrix per trial.

C. Lindblad dynamics

The density matrix $\hat{\rho}$ evolves under the Lindblad master equation

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \sum_{i=1}^{L} \gamma_i \Big(\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \tfrac{1}{2}\{\hat{L}_i^\dagger \hat{L}_i,\, \hat{\rho}\} \Big), \qquad (2)$$

where $\hat{L}_i = \hat{n}_i$ is a dephasing jump operator and γ_i is the dephasing rate at site i. Baseline dephasing is uniform: $\gamma_i = \gamma = 0.1$ for all sites. Since $\hat{L}_i = \hat{n}_i$ commutes with the total number operator $\hat{N} = \sum_i \hat{n}_i$, the dissipator decoheres number-state superpositions without changing any local expectation value $\langle \hat{n}_i \rangle$; both local and total particle number are conserved at all times.

The Lindblad equation is converted to a superoperator acting on the vectorised density matrix in the row-major convention. Time evolution is computed by the action of the matrix exponential on the state vector, using `scipy.sparse.linalg.expm_multiply` [16], which avoids forming the full propagator. No Trotter decomposition, stochastic unravelling, or other approximation is used.

D. Burn-in state

The initial state is the ground state of $\hat{H}$, obtained by exact diagonalisation. This pure state is then evolved under uniform baseline dephasing for a burn-in period $t_{\text{burn}} = 5.0$ (in units of $\hbar/U$). The resulting mixed state $\hat{\rho}_{\text{burn}}$ serves as the starting point for all interventions within a given condition. The burn-in ensures that the state has developed spatially heterogeneous fluctuation structure under dissipation before any intervention is applied.

E. Handle definition

The handle variable is the local occupation variance

$$F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2, \qquad (3)$$

evaluated on the post-burn-in state $\hat{\rho}_{\text{burn}}$. This quantity identifies sites with larger number fluctuations. Because the Hamiltonian, the ground state, and the baseline Lindblad operators are all invariant under site reflection $i \leftrightarrow L-1-i$, the density matrix $\hat{\rho}(t)$ remains symmetric at all times, and consequently $F_i(t) = F_{L-1-i}(t)$ exactly. The F_i profile therefore peaks at the geometrically central sites and is smallest at the edges at every point in the evolution, including at t_{burn}. The top-k sites by F_i ($k = \lceil L/3 \rceil$) are selected for intervention. The selected set is fixed by the post-burn-in state and is therefore deterministic within each condition.

F. Intervention protocol

The **targeted intervention** applies additional dephasing $\gamma_{\text{extra}} = 0.5$ at the selected high-F_i sites, keeping all other sites at the baseline rate γ. The **random control** applies the same total extra budget ($k \times \gamma_{\text{extra}}$) to k randomly chosen sites. In both cases the total dissipative budget is identical. Because $\hat{L}_i = \hat{n}_i$ commutes with the total number operator, $\langle \hat{n}_i \rangle$ is unchanged at the moment of intervention by construction; the occupation profile is held fixed at the start of each arm.

The targeted intervention is deterministic: for a given condition, the post-burn-in state and the selected sites are the same, so the targeted evolution is computed once per (condition, τ) pair. Only the random arm varies across trials.

G. Target

The primary target is local future occupation loss at the intervention sites:

$$\mathcal{L}_{\text{local}}(\tau) = \sum_{i \in \mathcal{S}} \max\big(0,\ \langle \hat{n}_i \rangle_t - \langle \hat{n}_i \rangle_{t+\tau}\big), \qquad (4)$$

where $\mathcal{S}$ is the set of selected sites and τ is the measurement horizon. The $\max(\cdot, 0)$ clip counts only sites that *lose* occupation; gains at other sites do not offset local losses.

H. Design and statistical analysis

Four tunnelling-to-interaction ratios $J/U \in \{0.12, 0.20, 0.30, 0.40\}$ are tested across all chain lengths, spanning from deep Mott-like to intermediate superfluid-like regimes. To resolve the crossover boundary identified in the initial sweep, a secondary boundary sweep tests

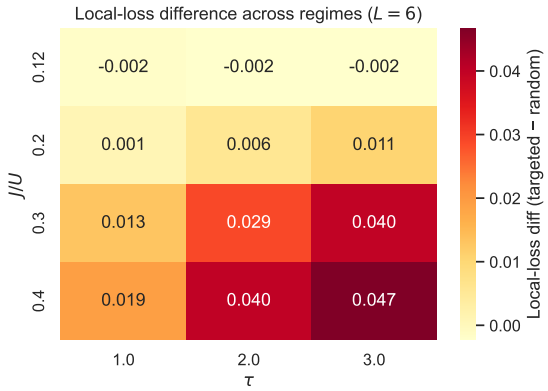

FIG. 1. Local-loss difference across J/U and τ at $L = 6$. The handle effect is positive in the intermediate and stronger coupling regime and absent or reversed at the lowest tested coupling.

$J/U \in \{0.16, 0.18, 0.20, 0.22, 0.24, 0.26, 0.28, 0.30\}$ at $L = 8$ and $J/U \in \{0.16, 0.18, 0.22, 0.24, 0.26, 0.28\}$ at $L = 9$, using horizons $\tau \in \{2, 3\}$. Three horizons $\tau \in \{1, 2, 3\}$ and four chain lengths $L \in \{6, 7, 8, 9\}$ are used in the primary sweep. For each condition, $N_{\text{trials}} = 100$ independent random-site draws are compared against the single targeted intervention. The test statistic is the mean difference in $\mathcal{L}_{\text{local}}$ between targeted and random interventions. A 95% bootstrap confidence interval (1000 resamples, trial-level resampling) is constructed for each condition. The decision rule is that a condition passes if the lower bound of the 95% CI exceeds zero.

III. RESULTS

A. Main result at $L = 6$

Table I reports the local-loss difference for all conditions at $L = 6$. At $L = 6$, the targeted intervention produces more local occupation loss than the random control throughout $J/U \in \{0.20, 0.30, 0.40\}$, with all confidence intervals in that range excluding zero, while the broader cross-size regime map (Sec. III C) shows that $J/U = 0.20$ is transitional. The lowest tested coupling $J/U = 0.12$ is uniformly negative: the targeted intervention produces *less* local loss than matched-budget random targeting at all three horizons.

The effect increases with J/U from the negative boundary at 0.12 through a transitional regime at 0.20 and into the robust positive pocket at 0.30–0.40. The magnitude of the positive difference also increases with the time horizon τ, consistent with a cumulative dynamical effect rather than a transient perturbation.

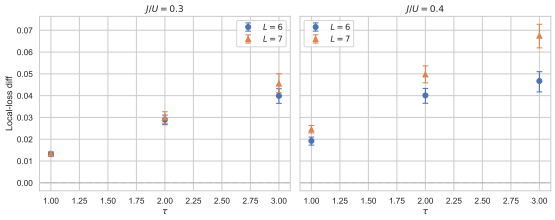

FIG. 2. Local-loss difference at $J/U = 0.30$ and 0.40 for $L = 6$ and $L = 7$ (two representative sizes). Error bars are 95% bootstrap CIs. The same positive result holds at $L = 8$ and $L = 9$; the positive pocket is robust across the full size ladder $L \in \{6, 7, 8, 9\}$.

B. Robustness across chain length

Table II shows pooled results at $J/U \in \{0.30, 0.40\}$ for $L \in \{6, 7\}$. The positive pocket at $J/U \in \{0.30, 0.40\}$ passes at all four tested chain lengths $L \in \{6, 7, 8, 9\}$ with 95% confidence intervals excluding zero at every horizon $\tau \in \{1, 2, 3\}$, confirming that the positive pocket is not a small-size artefact.

The intermediate coupling $J/U = 0.20$ is strongly size-sensitive: it passes at $L = 6$ across all horizons, requires $\tau = 3$ at $L = 7$, passes only at $\tau \in \{2, 3\}$ at $L = 8$, and passes only at $\tau = 3$ at $L = 9$. This size-dependent behaviour places $J/U = 0.20$ squarely in the crossover band rather than the robust positive pocket (see Sec. III D).

C. Regime map

The results define three regimes within the tested parameter range:

Negative regime ($J/U = 0.12$): The targeted intervention produces less local loss than random targeting at all four chain lengths and all horizons. In this deep Mott-like regime the occupation-variance handle does not identify causally effective sites; the 95% CIs are strictly negative at every (L, τ) combination tested.

Crossover band ($J/U \approx 0.18$–0.24, depending on L and τ): The handle begins to function but its effectiveness is size- and horizon-dependent. The first-pass onset shifts upward with L (for fixed τ) and downward with τ (for fixed L), both trends monotone across the ladder (see Sec. III D).

Positive pocket ($J/U \gtrsim 0.30$): The handle is consistently positive across all four chain lengths and all horizons $\tau \in \{1, 2, 3\}$. This is the regime where the occupation-variance selector reliably identifies sites whose perturbation produces excess local loss, with CIs strictly above zero at every tested (L, τ) combination.

The existence of a negative boundary regime and a structured crossover band is itself informative: the handle is not a generic property of the dephasing operator but is sharply regime-dependent.

TABLE I. Local-loss difference (targeted − random) at $L = 6$, $N_{\text{trials}} = 100$ per condition. Format: mean [95% CI]. $J/U = 0.12$ is uniformly negative (negative regime); $J/U = 0.20$ is transitional (crossover band); $J/U \in \{0.30, 0.40\}$ are robustly positive.

J/U	$\tau = 1$	$\tau = 2$	$\tau = 3$
0.12	−0.002 [−0.002, −0.001]	−0.002 [−0.003, −0.002]	−0.002 [−0.003, −0.001]
0.20	0.001 [0.000, 0.001]	0.006 [0.005, 0.007]	0.011 [0.009, 0.012]
0.30	0.013 [0.012, 0.014]	0.029 [0.027, 0.031]	0.040 [0.036, 0.043]
0.40	0.019 [0.017, 0.021]	0.040 [0.036, 0.043]	0.047 [0.042, 0.051]

TABLE II. Pooled local-loss difference for $L \in \{6, 7\}$ at $J/U = 0.30$ and 0.40 (two representative sizes from the full ladder). Format: mean [95% CI], pooling trial-level differences across $\tau \in \{1, 2, 3\}$. All four tested sizes $L \in \{6, 7, 8, 9\}$ pass at both couplings (see text).

Condition	$L = 6$	$L = 7$
$J/U = 0.30$	0.027 [0.025, 0.029]	0.030 [0.027, 0.032]
$J/U = 0.40$	0.035 [0.032, 0.038]	0.047 [0.044, 0.051]

TABLE III. First-PASS onset coupling: smallest J/U at which the 95% CI strictly excludes zero, by chain length L and horizon τ. Entries ≤ 0.20 are bounded by the initial sweep (exact onset lies in the interval (0.12, 0.20]); entry > 0.20 indicates a failure at $J/U = 0.20$ with onset in (0.20, 0.30) (not resolved by the boundary sweep, which focused on $L \geq 8$).

L	$\tau = 2$	$\tau = 3$
6	≤ 0.20	≤ 0.20
7	> 0.20	≤ 0.20
8	0.20	0.18
9	0.24	0.20

D. Crossover-band onset

The boundary sweep resolves the smallest J/U at which the 95% CI first strictly excludes zero (first PASS) for $L \in \{8, 9\}$. Table III reports these onset couplings for horizons $\tau \in \{2, 3\}$.

The onset coupling is monotonically increasing with L for fixed τ, and monotonically decreasing with τ for fixed L, across the full tested range. The convergence of these two trends in the range $J/U \approx 0.18$–0.24 defines the empirical crossover band reported in the abstract. In particular, any coupling in $J/U \gtrsim 0.30$ lies above the onset for all tested (L, τ) combinations, consistently defining the positive pocket boundary.

E. Spatial response pattern

Fig. 3 shows the site-resolved occupation change under targeted versus random intervention at $J/U = 0.30$, $L = 6$, $\tau = 2$. The bars show the mean site-resolved occupation change under the deterministic targeted intervention and averaged across the 100 matched-budget random interventions.

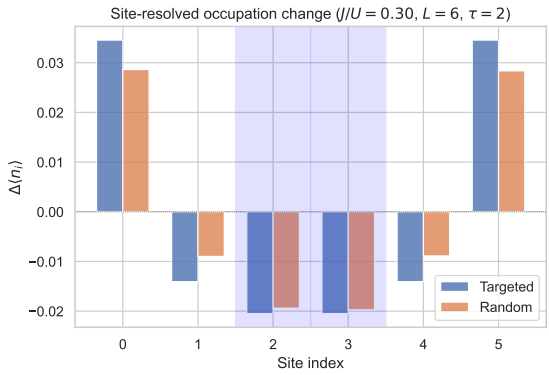

FIG. 3. Mean site-resolved occupation change under the targeted intervention and across matched-budget random interventions ($J/U = 0.30$, $L = 6$, $\tau = 2$). Negative values correspond to occupation loss. Shaded bands mark selected sites. The spatial response is consistent with nonlocal redistribution.

The observed pattern is consistent with a nonlocal, redistributive response. Under targeted intervention, the selected sites (shaded) tend to show *less* negative occupation change than under random intervention: selected-site loss is reduced relative to random targeting. At the same time, non-selected sites show increased occupation loss under targeted intervention. The net effect is that the intervention changes the spatial pattern of occupation change across the chain rather than simply amplifying loss at the targeted sites themselves.

This is the opposite of what a purely local-instability mechanism would predict, in which extra dephasing at a high-F_i site would increase loss specifically at that site. The data are more consistent with a picture in which perturbing high-F_i sites alters the subsequent spatial distribution of occupation change across the chain.

IV. DISCUSSION

a. Summary. In an open Bose–Hubbard chain, sites of high local occupation variance serve as causal handles in the intermediate and stronger coupling regime: targeting them with additional dephasing produces more local future occupation loss than matched-budget random targeting at $J/U \in \{0.30, 0.40\}$, robustly across the

full tested size ladder $L \in \{6, 7, 8, 9\}$ and all tested horizons $\tau \in \{1, 2, 3\}$. The effect is absent or reversed in the negative regime ($J/U = 0.12$) and size- and horizon-dependent in the crossover band $J/U \approx 0.18$–0.24, with the first-pass onset shifting upward with L and downward with τ, both trends monotone across the ladder.

b. Mechanism. The site-resolved response is more consistent with nonlocal redistribution than with local instability. Perturbing high-F_i sites is associated with a redistribution of subsequent occupation change across the chain. A complete mechanistic account is beyond the scope of the present analysis.

c. What is not claimed. We do not claim that high-F_i targeting is optimal among all possible intervention strategies; that the effect extends to all coupling regimes (it does not hold at $J/U = 0.12$ and is transitional in the crossover band $J/U \approx 0.18$–0.24); that the mechanism generalises to other quantum systems or to chains beyond the tested range; or that the occupation-variance handle corresponds to any equilibrium observable.

d. Scope and limitations. The causal-handle effect is regime- and size-dependent in the present design, with the clearest fully robust support in the $J/U = 0.30$–0.40 regime. The Hilbert-space dimension of the fixed-N sector grows combinatorially with L and $n_{\max}$, limiting exact Lindblad evolution to small chains. The tested range ($L \leq 9$, $n_{\max} = 3$) establishes the effect in the positive pocket and sharpens the crossover band to $J/U \approx 0.18$–0.24 (depending on system size and horizon), but does not access the thermodynamic limit. Whether the positive pocket persists, broadens, or shifts at larger system sizes is an open question requiring approximate methods such as matrix-product-operator Lindblad evolution [14]. A further structural limitation of the present design is that, due to the reflection symmetry established above, the high-F_i selector and a geometric central-k selector are identical in this model. The experiment therefore establishes that targeting the k central sites produces more occupation loss than matched-budget random targeting in the positive pocket; it cannot presently attribute the effect to the variance information specifically rather than to the spatial position. Disentangling the two requires breaking the spatial symmetry — through, for example, disorder, asymmetric filling, or site-dependent coupling — and is left for future work.

e. Experimental relevance. The 1D open Bose–Hubbard model under dephasing is experimentally motivated. Cold-atom optical-lattice platforms already realise 1D Bose–Hubbard-like systems with tunable J/U [2, 4], and recent advances in quantum-gas microscopy provide site-resolved preparation and readout relevant to the targeted-intervention protocol studied here [17]. In particular, groups working with 1D optical lattices have demonstrated single-site addressing and out-of-equilibrium dynamics in regimes that overlap with the coupling range explored here. The Lindblad dephasing modelled by $\gamma \hat{n}_i$ is a standard approximation for photon-scattering-induced decoherence in optical lattices [9, 14]. The present results therefore constitute a theory-side proposal with experimental plausibility: the positive-pocket and crossover-band structures identified here are, in principle, accessible to existing site-resolved cold-atom platforms, though the exact causal-handle protocol has not yet been implemented experimentally.

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DATA AND CODE AVAILABILITY

All simulation code and raw data are available at https://github.com/kunalb541/BH.

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